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THE LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE

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Table of Contents

Part 1 3

1. Random walk Metropolis simulation 3

2. Convergence assessment using RbRb 3

3. Conclusion 4

Part 2 5

2 (a) Best times and days of the week to minimise delays 5

2(b) Do older planes suffer more delays on a year-to-year basis? 6

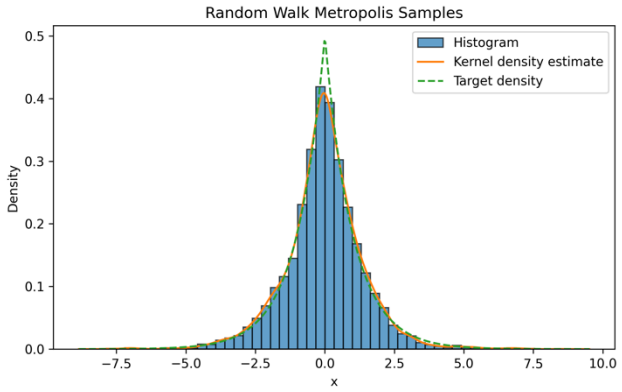
2(c) Logistic regression model for diverted flights 8

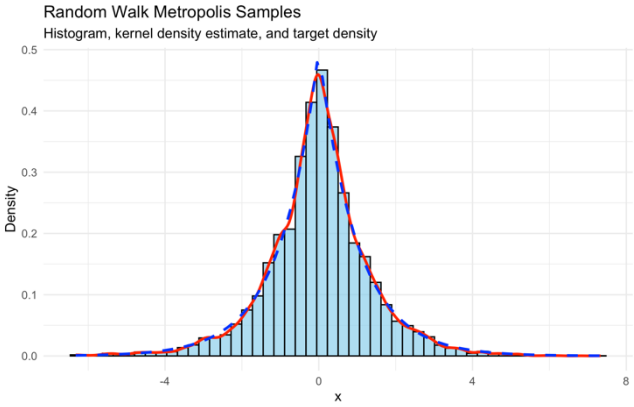
Part 1

1. Random walk Metropolis simulation

The aim of this part was to simulate random values from the target density $f(x) = \frac{1}{2}\exp(-|x|)$, defined over the *real line*, using the random walk Metropolis algorithm. The analysis was carried out independently in both R and Python so that the two implementations could be compared. In each case the algorithm was run with N = 10000 iterations, proposal standard deviation s = 1, and initial value $x_0 = 0$. At each step, a candidate value was drawn from a Normal distribution centred at the current state, and the move was accepted or rejected according to the Metropolis rule. To avoid numerical overflow, all acceptance comparisons were performed on the log scale.

The Python implementation produced a sample mean of **−0.0280** and a sample standard deviation of **1.4067**. These are consistent with the theoretical properties of the target distribution, which has mean 0 and is symmetric around it. The R implementation produced a sample mean of **-0.0052** and a sample standard deviation of **1.3442** . Although the exact values differ slightly between languages due to their independent random number streams, both results point to the same conclusion: the algorithm is sampling successfully from the required distribution.





In both figures, the histogram and kernel density estimate closely follow the overlaid curve of f(x), with the characteristic double-exponential (Laplace) shape visible in all three. This agreement between the simulated distribution and the target density confirms that the algorithm was implemented correctly in both languages and that the generated samples are suitable for **Monte Carlo estimation**.

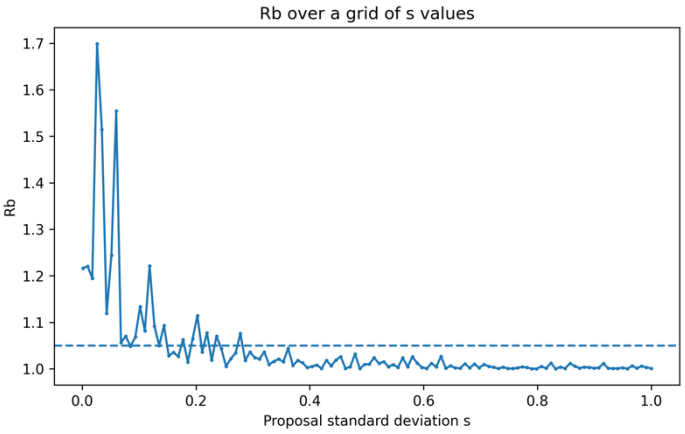
2. Convergence assessment using R̂

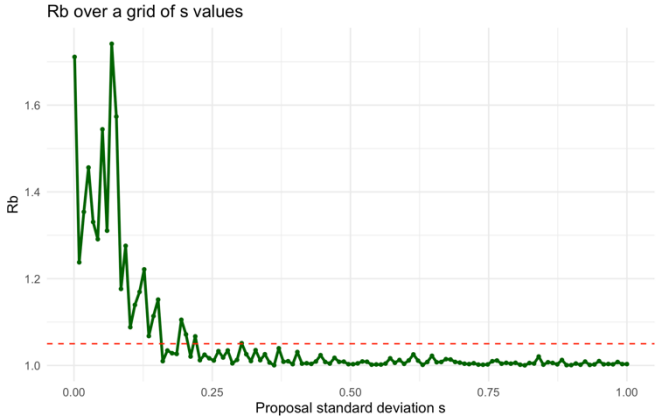
To assess whether the algorithm had converged, the Gelman–Rubin diagnostic R̂ was computed. Four chains were generated starting from the same initial value $x_0 = 0$, and R̂ was calculated as:

$$\hat{R} = \sqrt{\frac{B+W}{W}}$$

where W is the average within-chain variance and B is the between-chain variance across the J = 4 chains. Values of $\hat{R}$ close to 1 indicate that the chains are behaving similarly and have likely converged, while values above 1.05 are generally taken as evidence of poor mixing.

Using the required settings of N = 2000, s = 0.001, and J = 4, the Python implementation gave an $\hat{R}$ value of **1.2162**, and the R implementation gave a comparable value. Both are above the 1.05 threshold, which indicates that convergence is not fully achieved under this proposal scale. This is physically intuitive: when s = 0.001 the proposal distribution is extremely narrow, so the chains move only in tiny increments and explore the state space very slowly. Even over 2000 iterations the chains *do not mix well*, and this is captured by the elevated $\hat{R}$.





The plots of $\hat{R}$ across a grid of s values from 0.001 to 1 make this relationship clear. At the smallest values of s the diagnostic is noticeably elevated, reflecting the poor mixing caused by overly cautious proposals. As s increases, the chains are able to move more freely through the distribution, and $\hat{R}$ falls toward 1, eventually dropping below the 1.05 threshold and staying there for larger values of s. This confirms that the choice of proposal standard deviation has a *substantial effect* on the practical performance of the **random walk Metropolis algorithm.**

3. Conclusion

Both the R and Python implementations produced consistent simulation results, with generated samples that closely matched the target density in shape and whose Monte Carlo estimates of the mean were close to the theoretical value of zero. The convergence study showed that a very small proposal scale of s = 0.001 leads to poor chain mixing and $\hat{R}$ values above the acceptable threshold. As s increases, convergence *improves steadily* and the diagnostic drops to near 1. The algorithm itself is correct in both implementations; its practical behaviour depends heavily on the choice of s.

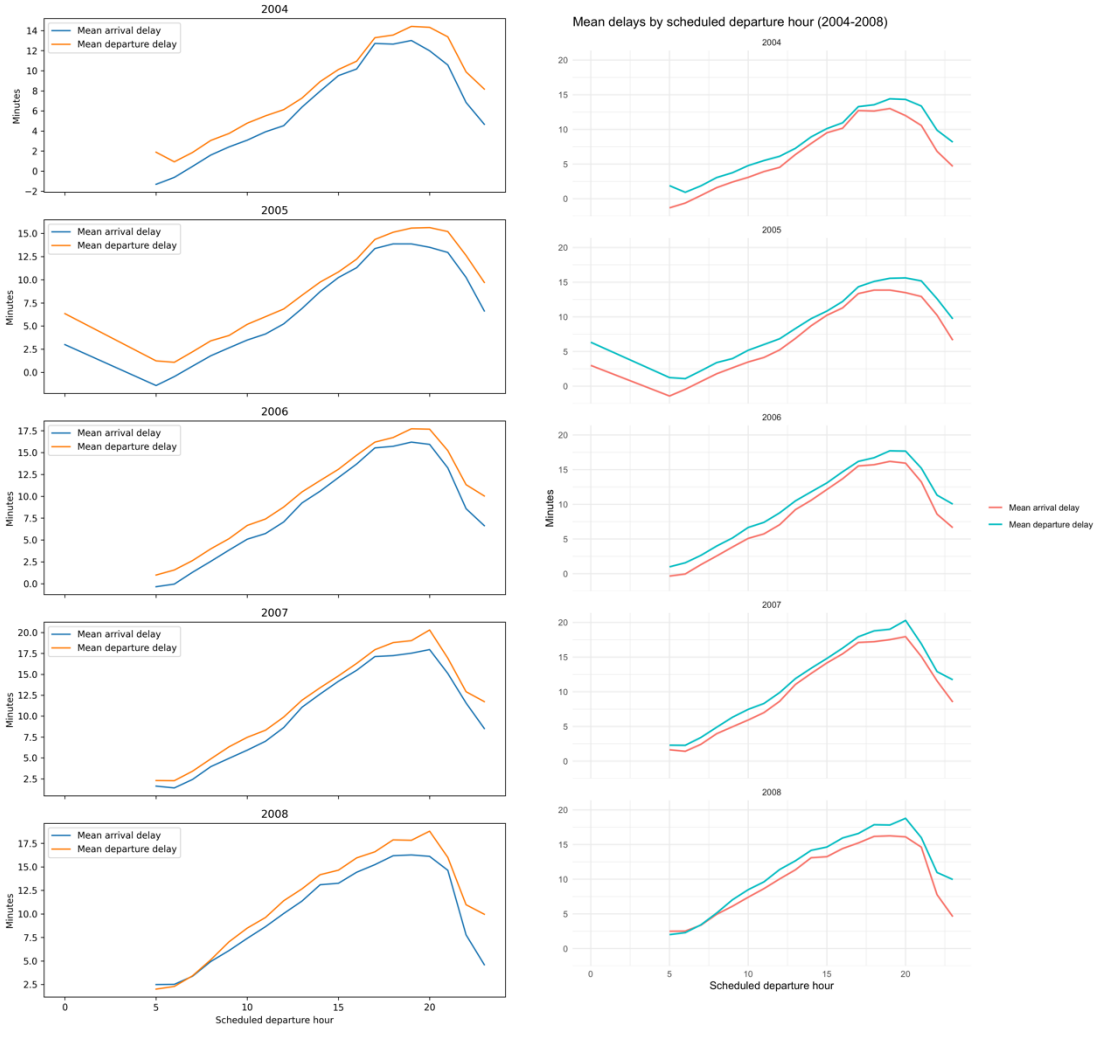
Part 2

The dataset used throughout Part 2 covers commercial domestic US flights from **2004 to 2008**, sourced from the 2009 ASA Statistical Computing and Graphics Data Expo via the Harvard Dataverse. After combining the five yearly files, the raw dataset contained approximately **31.25 million** flight records across **19 variables**. All analysis was carried out in Python with reason to computational constraints, with the results mirrored in R using pre-saved summary outputs.

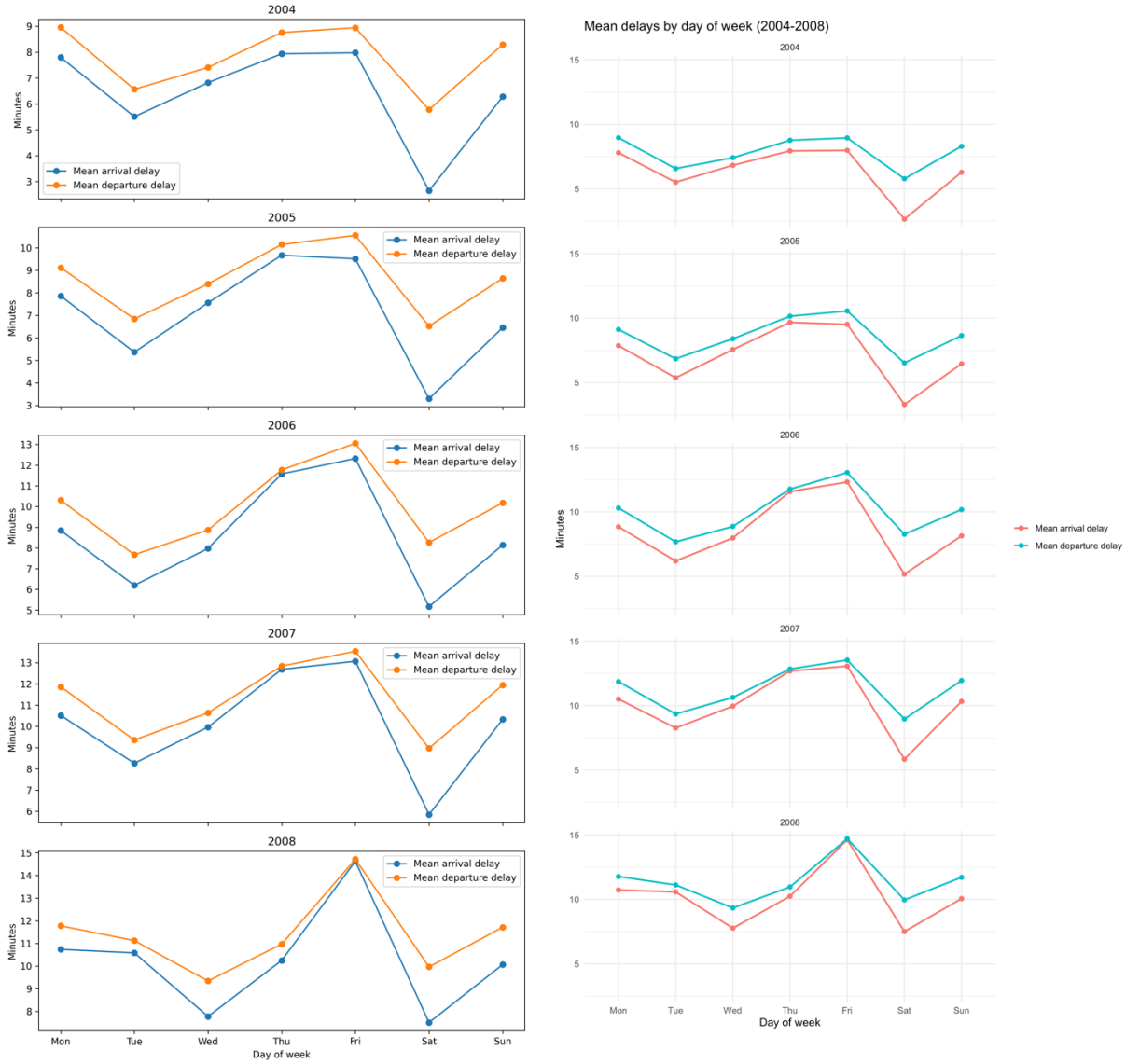
2(a) Best times and days of the week to minimise delays

To identify the best departure times and days, flights were grouped by year, scheduled departure hour, and day of week. Cancelled flights were excluded throughout, as their recorded delay values are not meaningful. For each group, the mean arrival delay and mean departure delay were calculated. Because some low-traffic hours produced unreliable averages based on only a handful of flights, a minimum threshold of 10,000 flights per hour-year combination was applied before identifying the best departure hour.

After applying this filter, the **best scheduled departure hour** in each year was consistently **05:00**, except in 2007 where it was **06:00**. These early morning departures had the lowest mean arrival and departure delays across all five years, which aligns with the well-known operational pattern where early flights benefit from a clean slate before delay accumulation builds up through the day. Without the filter, the apparent "best" hours were extreme outliers with very low flight counts, for example, a handful of flights at 04:00 in 2004, which illustrates why the threshold is necessary for a meaningful answer.



The day-of-week analysis showed an even more consistent pattern. **Saturday** had the lowest mean arrival delay in every single year from 2004 to 2008, making it the most reliable day to fly for minimising delays across the entire period. In contrast, Thursday and Friday consistently recorded the highest average delays, particularly in later years. The Saturday advantage likely reflects lower overall traffic volumes compared with the peak business travel days midweek and on Fridays.



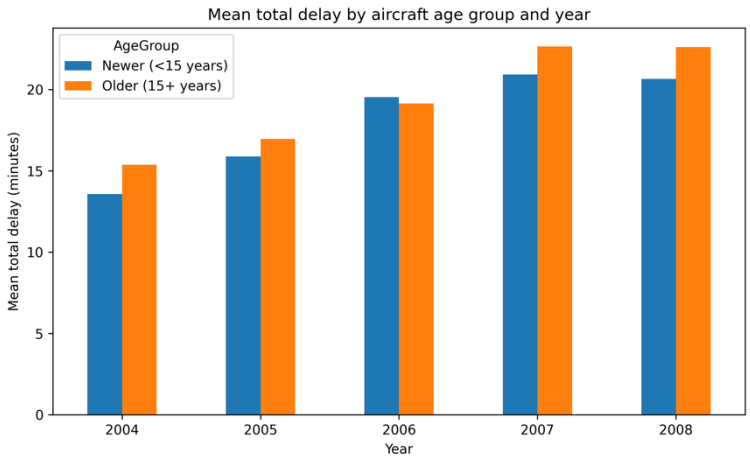
In summary, the data consistently points to **early morning departures** (05:00–06:00) and **Saturdays** as the best combination for minimising delays over the 2004–2008 period.

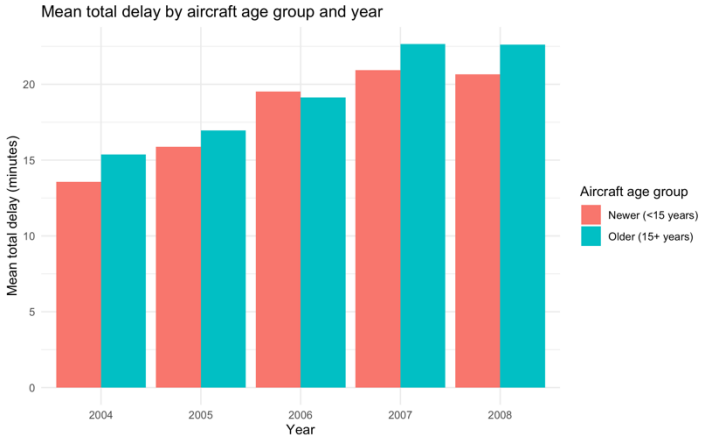
2(b) Do older planes suffer more delays on a year-to-year basis?

To examine whether aircraft age is associated with greater delays, the flight records were merged with manufacturing year information from the supplementary plane-data file using the aircraft tail number. Aircraft age for each flight was then calculated as the difference between the flight year and the plane's manufacturing year. Observations with missing or implausible age values (negative ages or ages above 50 years) were excluded. A combined total delay measure, the sum of arrival delay and departure delay, was used as the main outcome.

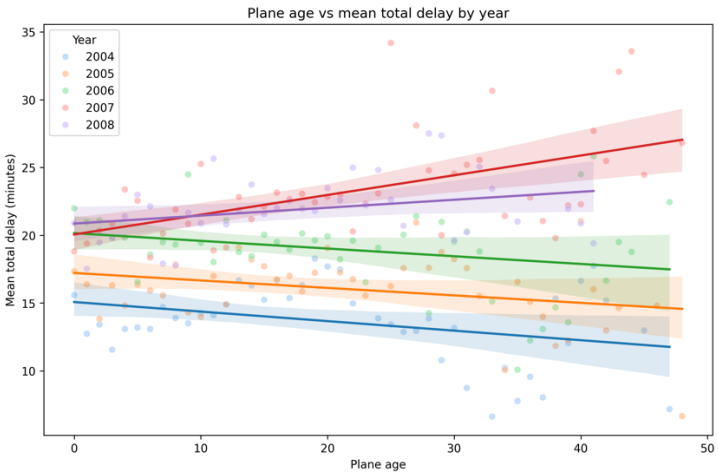
The match rate between flight records and the plane-data file was approximately **75%**, meaning around a quarter of flights could not be linked to a manufacturing year and were excluded from this part of the analysis.

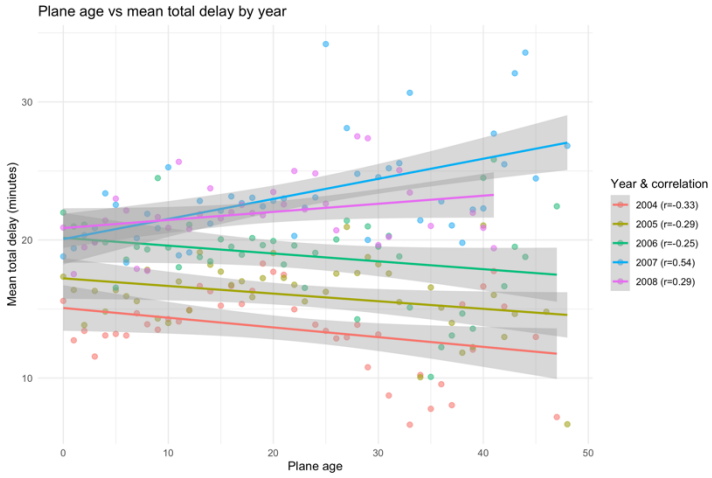
Planes were first split into two age groups: **newer** (under 15 years old) and **older** (15 years or more). In **2004, 2005, 2007, and 2008**, the older group recorded a higher mean total delay than the newer group. The differences were small but consistent in those years. However, **2006 was an exception**: newer planes averaged a mean total delay of approximately **19.53 minutes** compared with **19.13 minutes** for older planes, meaning newer planes actually had slightly higher average delays that year. This inconsistency suggests the relationship between age and delay is not straightforward.





Looking at exact aircraft ages rather than broad groups, the trend was similarly weak. The year-by-year correlations between plane age and total delay were very small, ranging from approximately −0.006 to +0.013, indicating *almost no linear relationship* between the two variables in any single year. In some years the slope of the age–delay relationship was positive, while in others it was slightly negative.



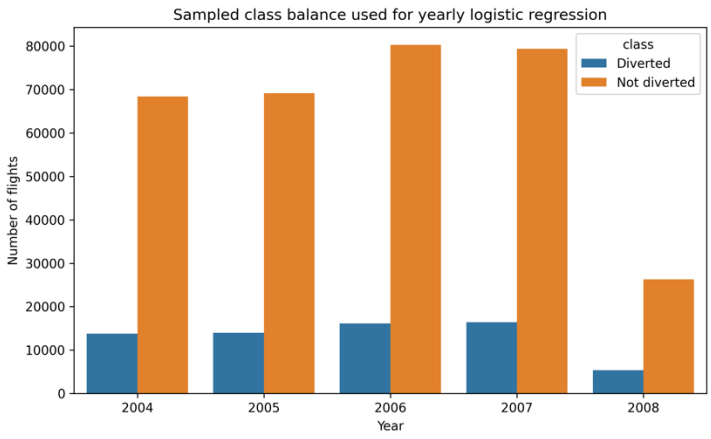


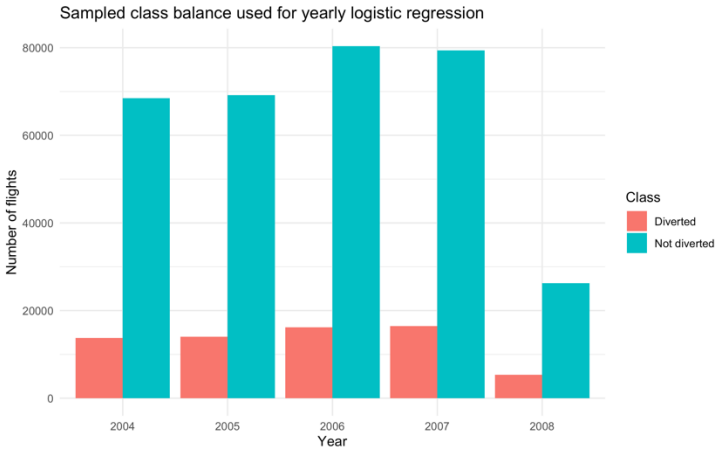
Overall, the evidence suggests that older planes experienced marginally higher average delays in most years, but the effect is small, inconsistent across years, and unlikely to be practically significant on its own. Aircraft age does not appear to be a strong standalone predictor of flight delay in this dataset.

2(c) Logistic regression model for diverted flights

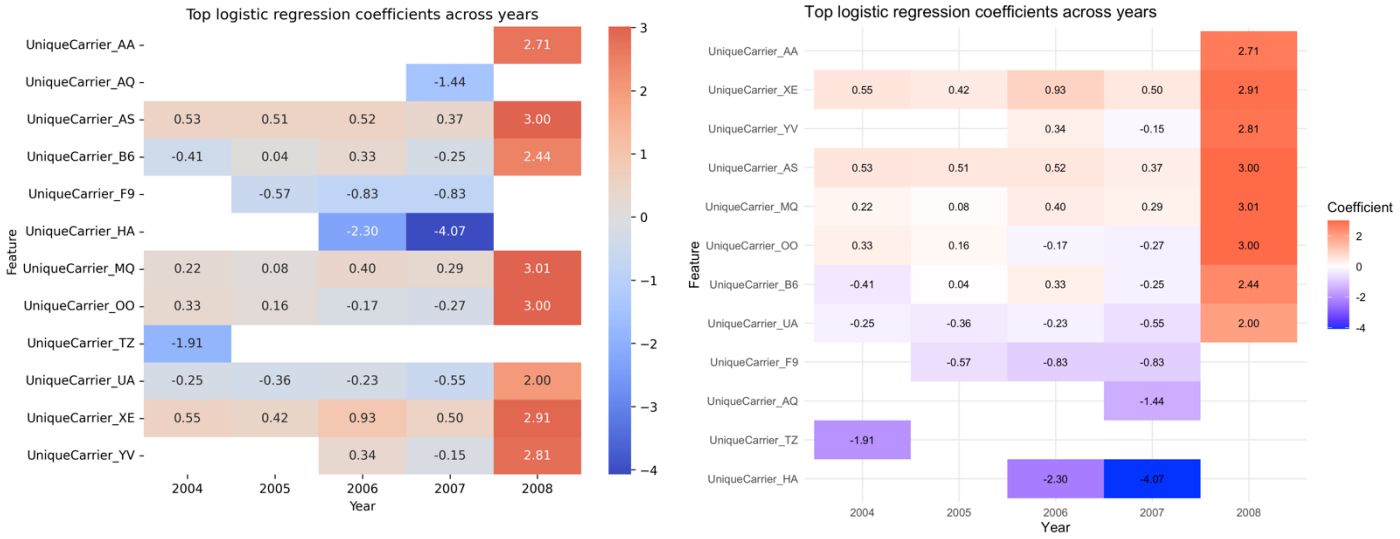
A separate logistic regression model was fitted for each year from 2004 to 2008 to model the probability of a flight being diverted. The predictor set was built from the variables specified in the coursework: departure date attributes (month, day of month, day of week), scheduled departure and arrival hours, distance, the coordinates of both the origin and destination airports, and carrier. Airport coordinates were merged in from the supplementary airports file using the IATA code. The carrier variable was encoded as a *set of indicator (dummy) variables.*

Data preparation and class imbalance. Diverted flights are extremely rare in this dataset, they represent well under 1% of all operated flights. Fitting logistic regression directly on the full dataset would be computationally prohibitive and would produce a model dominated by the non-diverted majority. To address this, the modelling dataset for each year was constructed by retaining all diverted flights and combining them with a random sample of non-diverted flights at a 5:1 ratio. This produced a more balanced and computationally tractable fitting sample while preserving the key structure of the problem. Carriers with fewer than 500 flights in the sampled dataset, or whose sampled flights were all of the same diversion class, were excluded before fitting to avoid complete separation problems that cause coefficient estimates to diverge.

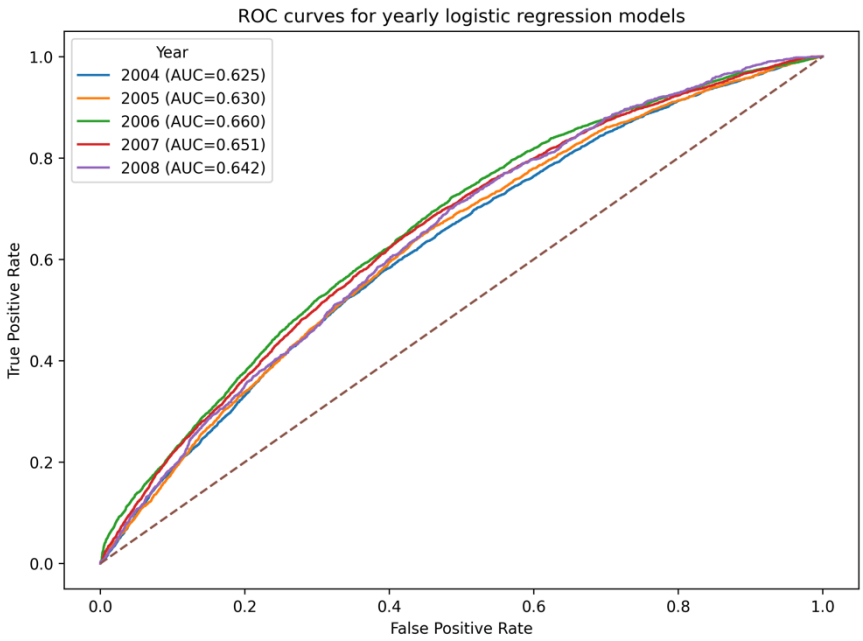


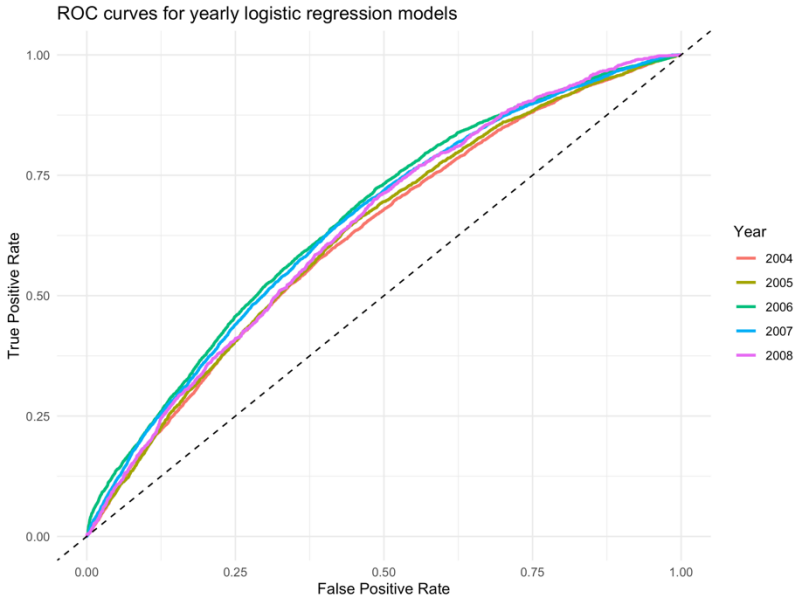


Model coefficients. The heatmap below shows the top features by average absolute coefficient magnitude across the five years. Carrier-related variables dominated the top of the list in most years, indicating that diversion risk differed substantially across airlines even after controlling for scheduling, distance, and route characteristics. Some carriers showed relatively stable coefficients across years, while others appeared only in certain years due to market entry, exit, or mergers during the 2004–2008 period, which itself is useful context for interpreting the year-to-year variation. Non-carrier features such as departure hour and arrival hour also contributed, though their individual coefficients were smaller.



Predictive performance. ROC curves were produced for each year using a 70/30 train–test split on the sampled modelling data. The AUC values were **0.6245** in 2004, **0.6303** in 2005, **0.6603** in 2006, **0.6506** in 2007, and **0.6424** in 2008. All five models performed better than random classification (AUC = 0.5), but the discrimination ability remained moderate rather than strong. The models improved noticeably between 2004 and 2006, before dipping slightly in 2008, the year with by far the fewest records due to the dataset only covering part of that year





The R outputs were produced from the same coefficient and ROC data saved by the Python pipeline and are consistent with the results reported above.

Conclusion

Across 2004 to 2008, the analyses point to several clear patterns. Early morning flights — particularly at 05:00 — and Saturday departures were consistently associated with the lowest average delays. Older aircraft tended to have slightly higher delays than newer ones in most years, but the relationship was weak and not consistent across all five years, with 2006 showing the reverse pattern.

The yearly logistic regression models demonstrated that diversion probability is partially explained by a combination of carrier identity, scheduled timing, distance, and airport location, though the overall predictive performance remained moderate, with AUC values ranging from 0.62 to 0.66. Taken together, the findings reflect the complex, multi-causal nature of flight delays and diversions, where no single variable dominates and the patterns shift meaningfully from year to year.

Note to Examiner: All charts/diagrams appended on the **Left-side** of the report have been generated from *python* and similarly, the charts/diagrams on the **Right-side** of the report have been generated from R.

The inner workings of how data was derived has only been briefly explained. The appended workbook files in the submission link has the complete process *in detail.*